CSIT751 (Core Java)

Module 2 – Java with Object Orientated Features

Programming Exercises for Practice (Practical Home Assignment 1)

1. Class & Object Basics

Problem Statement:
Create a Book class with title, author, and price as data members. Write methods to set and display details. Create two Book objects in main() and print their details.

Sample Input Format:

Java Basics
John Smith
500
Python Essentials
Mary John
650

Sample Output Format:

Title: Java Basics
Author: John Smith
Price: 500.0
Title: Python Essentials
Author: Mary John
Price: 650.0

---

2. Encapsulation

Problem Statement:
Create a BankAccount class with private members accountNumber and balance. Provide public methods to deposit, withdraw, and checkBalance using proper validation.

Sample Input Format:

12345
1000
deposit 500
withdraw 300

Sample Output Format:

Deposit successful. Current balance: 1500.0
Withdrawal successful. Remaining balance: 1200.0

---

3. Abstraction with Classes

Problem Statement:

Create an abstract class Shape with abstract methods area() and perimeter(). Extend it into Circle and Rectangle classes.

Sample Input Format:

circle
5
rectangle
4 6

Sample Output Format:

Circle Area: 78.54
Circle Perimeter: 31.42
Rectangle Area: 24.0
Rectangle Perimeter: 20.0

---

4. Method Overloading

Problem Statement:

Create a Calculator class with multiple add methods for different argument types.

Sample Input Format:

add 3 4
add 2.5 3.2
add 4 5 6

Sample Output Format:

Sum (int,int): 7
Sum (double,double): 5.7
Sum (int,int,int): 15

---

5. Constructors

Problem Statement:

Create a Student class with default, parameterized, and copy constructors. Print student details in each case.

Sample Input Format:

John 20

Sample Output Format:

Default Constructor: Name=Unknown, Age=0
Parameterized Constructor: Name=John, Age=20
Copy Constructor: Name=John, Age=20

---

6. Single Inheritance

Problem Statement:

Create a base class Person with attributes name and age. Derive a class Employee with empId and salary.

Sample Input Format:

Alice 30 E123 50000

Sample Output Format:

Name: Alice
Age: 30
Employee ID: E123
Salary: 50000.0

---

7. Multilevel Inheritance

Problem Statement:

Create Animal → Mammal → Dog classes. Show data and method access across multiple levels.

Sample Input Format:

Buddy Labrador

Sample Output Format:

Dog Name: Buddy
Breed: Labrador
Animal Type: Mammal

---

8. Method Overriding

Problem Statement:
Create a Vehicle class with a method start(). Override it in subclasses Car and Bike.

Sample Input Format:

car
bike

Sample Output Format:

Car starting...
Bike starting...

---

9. Role of Constructors in Inheritance

Problem Statement:
Show sequence of constructor calls in base and derived classes.

Sample Input Format:

create derived

Sample Output Format:

Base Constructor Called
Derived Constructor Called

---

10. Abstract Class & Method

Problem Statement:

Create abstract class Payment and extend it into CreditCardPayment and UPIPayment.

Sample Input Format:

creditcard 500
upi 300

Sample Output Format:

Processing credit card payment of 500
Credit card payment successful
Processing UPI payment of 300
UPI payment successful

---

11. Final Keyword

Problem Statement:

Demonstrate final class, final method, and final variables.

Sample Input Format:

test final

Sample Output Format:

Final variable value: 100
Cannot inherit from final class
Cannot override final method

---

12. Static Members

Problem Statement:

Create a class with a static variable to track the number of objects created.

Sample Input Format:

create 3 objects

Sample Output Format:

Number of objects created: 3

---

13. Super Keyword

Problem Statement:

Demonstrate using super to call parent constructor, method, and variable.

Sample Input Format:

test super

Sample Output Format:

Parent constructor called
Parent method called
Parent variable value: 50

---

14. Garbage Collection

Problem Statement:

Create a class with a finalize() method to display a message when an object is destroyed.

Sample Input Format:

create and destroy objects

Sample Output Format:

Object destroyed

---

15. Interface Implementation

Problem Statement:

Create interface Playable and implement it in Guitar and Piano classes.

Sample Input Format:

guitar
piano

Sample Output Format:

Playing Guitar...
Playing Piano...

---

16. Extending Interfaces

Problem Statement:

Create interfaces Readable and Writable and extend them into ReadableWritable.

Sample Input Format:

document read
document write

Sample Output Format:

Reading Document...
Writing Document...

---

17. Default & Static Methods in Interface

Problem Statement:

Create an interface with default and static methods.

Sample Input Format:

square
triangle

Sample Output Format:

Shape: Square
Shape: Triangle
Static method from interface

---

18. Interface Constants

Problem Statement:

Create an interface Game with constant MAX_PLAYERS.

Sample Input Format:

cricket
football

Sample Output Format:

Game: Cricket, Max Players: 11
Game: Football, Max Players: 11
